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STAFF REPORT FOR THE 2014 ARTICLE IV CONSULTATION, FIRST REVIEW UNDER THE THREE-YEAR ARRANGEMENT UNDER THE EXTENDED CREDIT FACILITY, AND REQUEST FOR WAIVER AND MODIFICATION OF PERFORMANCE CRITERIA—DEBT SUSTAINABILITY ANALYSIS

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This joint World Bank/IMF Debt Sustainability Analysis (DSA) has been prepared in the context of the First Review under the program supported by the IMF's Three-Year Extended Credit Arrangement (ECF). It is based on new end-2013 debt data. Based on lower WEO projections for gold prices and projected higher spending, the current account and fiscal balances are notably worse, with an impact on growth through the medium term. A downward adjustment to actual gold production statistics in 2012 and 2013 significantly worsened the current account balance, thus worsening the projection profile. A cautious assumption is made that gold mining reserves will begin facing depletion around 2022, not fully compensated by other mineral exports. Under these assumptions, none of the external debt ratios under the baseline scenario breach their respective indicative debt distress thresholds, but one of the standardized stress tests shows a breach, indicating that Burkina Faso's risk rating for debt distress remains moderate.[1] To supplement the standardized DSA treatment, staff has constructed an alternate scenario with a significant near term "growth shock." Under this scenario, the same standardized stress test shows an earlier breach in addition to a new breach under a different stress test.

[1] Burkina Faso's latest CPIA score is 3.8, which places the country's policy and institutional framework in the group of 'strong' performers.

BACKGROUND AND UNDERLYING DSA ASSUMPTIONS

1. Burkina Faso's nominal stock of debt as of end-2013 was 29.1 percent of GDP, equivalent to approximately US$3.6 billion (Table 1). Although up slightly from end-2012, this amount is lower than what had been assumed in the 2013 DSA (US$3.9 billion). The composition of debt shifted slightly more toward domestic debt, reflecting large issuances of government bonds in 2013, delayed donor disbursements, as well as exchange rate appreciation. About 75 percent of the total stock of debt at end-2013 was external debt, down from 81 percent at end-2012.

Table 1. Stock of Public Debt 2011–13
(CFAF billions)

	2011	2012	2013	Percent
Public Debt	1508.2	1616.3	1733.7	100
External Debt	1216.2	1290.6	1296.0	74.8
Domestic Debt	291.9	325.8	437.6	25.2
	(percent of GDP)			
	2011	**2012**	**2013**	
Public Debt	30.5	28.7	29.1	
External Debt	24.6	22.9	21.8	
Domestic Debt	5.9	5.8	7.4	

2. Table 2 and Box 1 summarize the main differences in macroeconomic assumptions between the previous DSA (July 2013) and this one. The most significant changes are gold price projections and their attendant impacts; retroactive revisions to gold production data by the authorities (Annex III, Box 1); and WEO projections for nominal exchange rate appreciation through the medium term. In the near term, somewhat lower growth is assumed, although this is harmonized with growth over the medium term. Maintaining spending roughly constant at current levels, these assumptions lead to higher fiscal and external deficits over the medium term. Moreover, the current account balance is expected to deteriorate further starting in 2022 when gold production slows, although there is modest improvement in fiscal and external balances in the longer run, as spending levels and imports adjust.

Table 2. Changes in Assumptions for the April 2014 DSA vs. the July 2013 DSA

		2013	2014	2015	2016	2017	2030	2031	2032
Gold (tons)	2014 DSA	38.9	40.5	42.5	44.6	46.9	60.0	60.0	60.0
	2013 DSA	45.4	48.6	52.0	54.6	57.3	90.8	92.6	93.6
Gold (USD/ounce)	2014 DSA (WEO)	1411	1327	1343	1370	1398	1487	1487	1487
	World Bank		*1250*	*1230*	*1216*	*1203*			
	2013 DSA	1418	1432	1451	1468	1491	1844	1873	1903
Exports of goods (% of GDP)	2014 DSA	20.8	19.1	18.3	17.9	17.2	15.6	15.7	15.7
	2013 DSA	25.0	25.2	25.1	24.2	23.6	21.9	21.8	21.6
Real GDP growth (y/y)	2014 DSA	6.6	6.7	6.8	7.0	6.8	6.0	6.0	6.0
	2013 DSA	7.0	7.0	7.0	6.9	6.8	6.0	6.0	6.0
Current account (% of GDP)	2014 DSA	-7.0	-7.2	-7.0	-7.0	-7.0	-7.4	-7.3	-7.3
	2013 DSA	-2.2	-2.2	-2.0	-3.3	-4.1	-4.7	-4.8	-4.9
Overall balance (% of GDP)	2014 DSA	-3.5	-3.1	-3.1	-3.0	-3.6	-6.1	-5.9	-5.9
	2013 DSA	-2.4	-2.4	-2.4	-2.1	-3.9	-3.1	-3.0	-2.9

3. This DSA assumes continued modest use of nonconcessional financing over the forecast horizon. The authorities have already signed contracts using about 60 percent of the program continuous PC on nonconcessional external debt, mainly for the Donsin airport project and a road connecting Dedougou to Tougan northwest of the capital. The authorities have requested an upward modifications of the program continuous PC on nonconcessional external debtby approximately 0.2 percent of GDP in order to finance two additional highway projects (Table a. and Box 2).

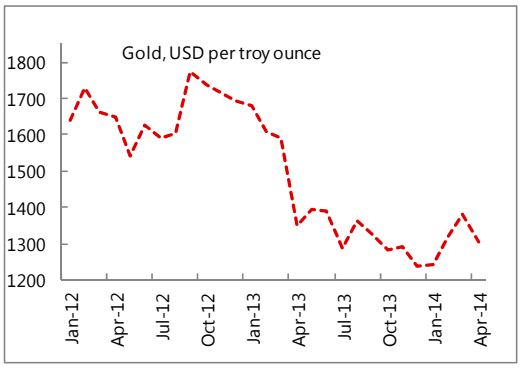
4. The DSA includes both already-contracted and anticipated borrowing on a disbursement basis. The authorities have reiterated their ongoing commitment to rely as much as possible on available concessional financing, but in light of limited concessional resources, this DSA includes an assumption that nonconcessional borrowing will be continued, at modest levels, through the DSA horizon. Consistent with this and the conservative assumption of less concessional financing going forward, the grant element of new borrowing is assumed to decrease gradually over the forecast horizon, especially toward the outer years.

Box 1. Burkina Faso: Macroeconomic Assumptions Underlying the DSA

Gold prices have dropped significantly since the last DSA. WEO projections have dropped as well, by an average of 20 percent over the forecast horizon.

Gold production is expected to rise steadily over the medium term with discovery of new reserves. The DSA assumes reserves will be gradually depleted after 2022.

GDP growth assumptions are lower in the medium term than in the 2013 DSA, based on the 2013 outturn, the near term outlook, and terms of trade projections. The longer term average remains unchanged at around 6 percent.



The overall fiscal deficit (including grants) is based on relatively stable, but higher spending levels, and revenues that fluctuate in line with gold exports. The deficit increases after gold production slows beyond 2022 and expenditures remain unchanged. In the long run, the deficit begins to improve as spending adjusts.

Debt composition: Domestic debt is assumed to increase modestly through the forecast horizon, reflecting the authorities' efforts to deepen domestic and regional financial markets. The remainder of the deficit is financed via external debt, but on less generous terms to reflect some nonconcessional financing and a more cautious assumption about ability to borrow at concessional terms going forward (the impact on the net present value is partly offset by use of a higher discount rate, as explained in paragraph 4).

The current account deficit is significantly larger throughout the forecast horizon than in the 2013 DSA, due to the authorities' revisions to gold production data for 2012 and 2013. This, plus gold price projections, render export projections significantly lower in the current DSA compared to the 2013 DSA. The current account deficit forecast mimics that of the fiscal balance, with a deterioration following an assumption of gradual depletion of gold reserves, with a delay in the adjustment of import levels.

Table a. Non-Concessional Borrowing

Project	Amount in bln of CFAF	Creditor
Ouagadougou Donsin International Airport	55	Islamic Bank of Development & ECOWAS Bank for Investment and Development
Dedougou-Tougan Highway Project	28.2	Islamic Bank of Development
Guiba-Garango Highway Project	25.7	In process
Kantchari-Diapaga-Tansarga (Benin border) Highway Project	38.8	In process

Box 2. Burkina Faso: Nonconcessional Borrowing

The nonconcessional borrowing for which approximately 83.2 bln CFAF has already been signed relates mainly to the Donsin Airport project and the highway project between Dedougou and Tougan. The remaining 64.5 bln CFAF that has not been signed is envisioned for two additional highway projects. Of this 64.5 bln CFAF, 12 billion is upward adjustment requested to the continuous PC on nonconcessional external debt.

The new airport project involves the revamping of the existing one from the city proper to 35 kilometers northwest of Ouagadougou. A World Bank feasibility study determined that the envisioned benefits would outweigh the costs, and supported its construction. Currently, two loans have been signed relating to the project. The first one is by the amount of 50 bln CFAF, financed by the Islamic Development Bank (BID) at an interest rate of 2.5 percent (grant element of 22 percent); this has been ratified by the National Assembly in March of this year. The second loan, in the amount of 5 bln CFAF, financed by ECOWAS, at an interest rate of 3 percent (grant element of 23 percent), is expected to be official soon.

The remaining loans relate to three highway projects that involve the bituminous surface treatment (BST) of existing roads or construction of new roads with BST treatment to improve road access. BST is usually used on low-traffic roads, but also as a type of sealing to rejuvenate an existing asphalt pavement. The first project for which 28.2 bln CFAF has been signed with BID at an interest rate of 2.5 percent (grant element of 20 percent) is a highway connecting Dedougou to Tougan northwest of the capital, approximately 92 km of distance. The loans for the remaining projects are estimated to be 25.7 and 38.8 CFAF bln respectively, but the financing has not yet been secured. The first project is a construction of a new road between Guiba and Garango, south-east of the country. The second is the construction of two connected highways that eventually connect Kantchari, located to the east of Ouagadougou near the Niger Border, to Tansarga, near the Benin border, involving approximately 70 km of paved road.

DSA RESULTS

A. External Debt

5. The 2014 DSA results are broadly unchanged from the 2013 exercise, reflecting a moderate risk of debt distress (Figure 2 and Table 1). The more pessimistic external and fiscal balances are offset by the higher discount rate, lower starting nominal external debt levels than had been assumed in the 2013 exercise, and revised assumptions about the mix of domestic and external financing. In the baseline scenario, all debt ratios continue to remain below the risk thresholds. Even with the less optimistic fiscal and external outlooks and the assumption for continued nonconcessional borrowing, the PV of external debt to exports peaked at 174.4 percent in 2034, still below the 200 percent debt risk threshold.[2]

[2] Residual figures in the external debt baseline are higher than in the 2013 DSA on account of the much larger current account deficit in 2012 and 2013 caused by the revision in official gold production data while actual fiscal results and debt accumulation remained unchanged. This discrepancy between the accrual of debt from the fiscal and estimated debt creating flows from the external balance likely reflects the underestimation of artisanal gold production/exports in the official data, as discussed in Box 1 of Annex III.

6. As in the past, the risk threshold for the PV of debt to exports is breached only in the case of one of the standardized stress tests. The breach is similar to that of the 2013 DSA. The stress test reflects a shock to exports, assuming that the growth of exports is reduced by one standard deviation in 2015–16. Indeed, volatility of gold prices remains an important vulnerability to exports. Under this shock scenario, exports are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock, implicitly assuming an offsetting adjustment in imports.

B. Total Public Debt DSA Results

7. The current DSA assumes a continued modest increase in domestic financing, bringing debt levels higher relative to the 2013 DSA, although the debt to GDP ratio remains below the relevant WAEMU convergence criterion. This assumption stems from progress already underway in expanding the domestic bond market, and plans to expand the regional market and develop more secondary markets. Under a fairly conservative assumption, this DSA assumes close to current terms for new domestic financing, which have fixed bid schemes and may be overly generous given the large oversubscription of recent bond issuances. However, the authorities are already considering variable bid schemes and other means to bring down the costs of domestic financing, and allow more precision for the amounts and timing thereof.

C. An Alternate Lower Growth Scenario

8. To challenge the robustness of the macroeconomic framework projections and the DSA results, staff constructed an alternative scenario. Since a gold price shock was explored in the 2012 and 2013 DSAs and is embodied in the standard stress tests, staff constructed an alternative scenario to test the sensitivity of the analysis to a near term shock to growth, stemming from a drought, but without recourse to additional budget support. In effect, this shock simply exaggerates the existing weather/growth patterns seen in the past. The shock is assumed to hit in the second half of 2014, reducing projected growth from 6.7 to 6.0 percent. The larger impact of the shock is felt in 2015, with projected growth slashed to 4.5 percent (from 6.8 percent). Public spending in 2015 would increase sharply to address food security needs and farm gate price subsidies to cotton farmers in addition to other spending aimed at muting the impact of the shock. At the same time, fiscal revenues would be affected by lower growth, causing a sharp increase in the deficit to 5.2 percent of GDP in 2015 (compared to 3.1 percent in the baseline), taking into account offsetting spending adjustment of 0.5 percent of GDP affecting domestically-financed investment.

9. Without additional budget support, the remaining fiscal deficit would be met by higher domestic financing and more expensive external financing. Of the 2.1 percentage points of GDP increase in the deficit, approximately 10 percent would be financed by additional domestic financing, and the rest would be met by additional nonconcessional external financing. Under this scenario, lower investment spending would be assumed to have an impact on growth and the fiscal deficit over the next 3 years, leading to higher accrual of debt. The current account deteriorates less relative to the baseline, as exports are assumed unchanged (with mining unaffected and cotton

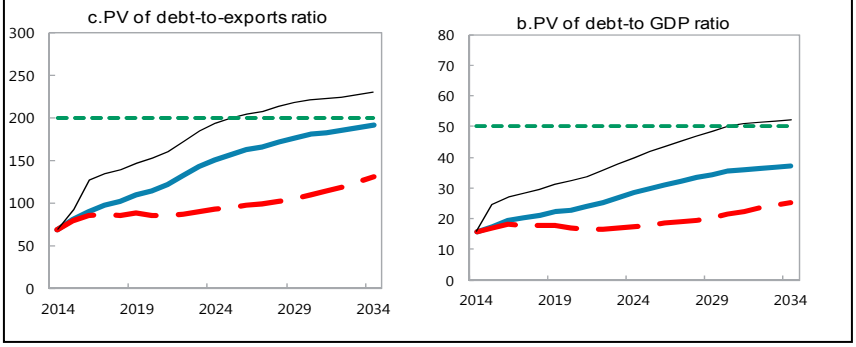
affected only marginally due to irrigation and GMO seeds) and lower growth and investment is assumed to have a dampening effect on import demand. The use of more nonconcessional external financing was proxied in the DSA framework by reducing the overall grant element of new financing for the medium term down to an approximate average of 10 percent.

10. DSA results under this scenario show a higher risk of debt distress, albeit with the risk rating remaining "moderate." On its own this scenario causes a near breach in the PV of debt-to-exports and in combining it with standardized stress tests, both PV of debt-to-exports and debt-to-GDP thresholds are breached. These results, even maintaining gold unchanged, indicate a clearer "moderate" risk of debt distress than the baseline, which is on the cusp of "moderate" and "low." However, all DSA results should be interpreted with caution, particularly given uncertain assumptions about the investment-growth nexus, outdated national account statistics, and balance of payments data still under revision.

Impact on Main Macroeconomic Aggregates of Weather Related Exogenous Shock

	2014	2015	2016	2017	2018	2019	2020
			(Percent of GDP)				
Real GDP Growth							
Downside Scenario	6.0	4.5	6.0	6.4	6.4	6.4	6.4
Baseline	6.7	6.8	7.0	6.8	6.7	6.6	6.4
Fiscal Deficit (including grants)							
Downside Scenario	-3.4	-5.2	-5.1	-4.4	-4.7	-4.9	-4.3
Baseline	-3.1	-3.1	-3.0	-3.6	-3.8	-4.0	-4.0
Current Account Balance							
Downside Scenario	-7.6	-7.5	-7.6	-8.3	-8.4	-8.4	-8.9
Baseline	-7.2	-7.0	-7.0	-7.0	-7.1	-7.2	-7.5
Debt/GDP Ratio							
Downside Scenario	22.7	24.0	25.3	25.6	26.2	26.9	27.4
Baseline	22.2	21.9	21.8	21.7	21.8	22.1	22.7

Figure 1. Alternative Low Growth Scenario



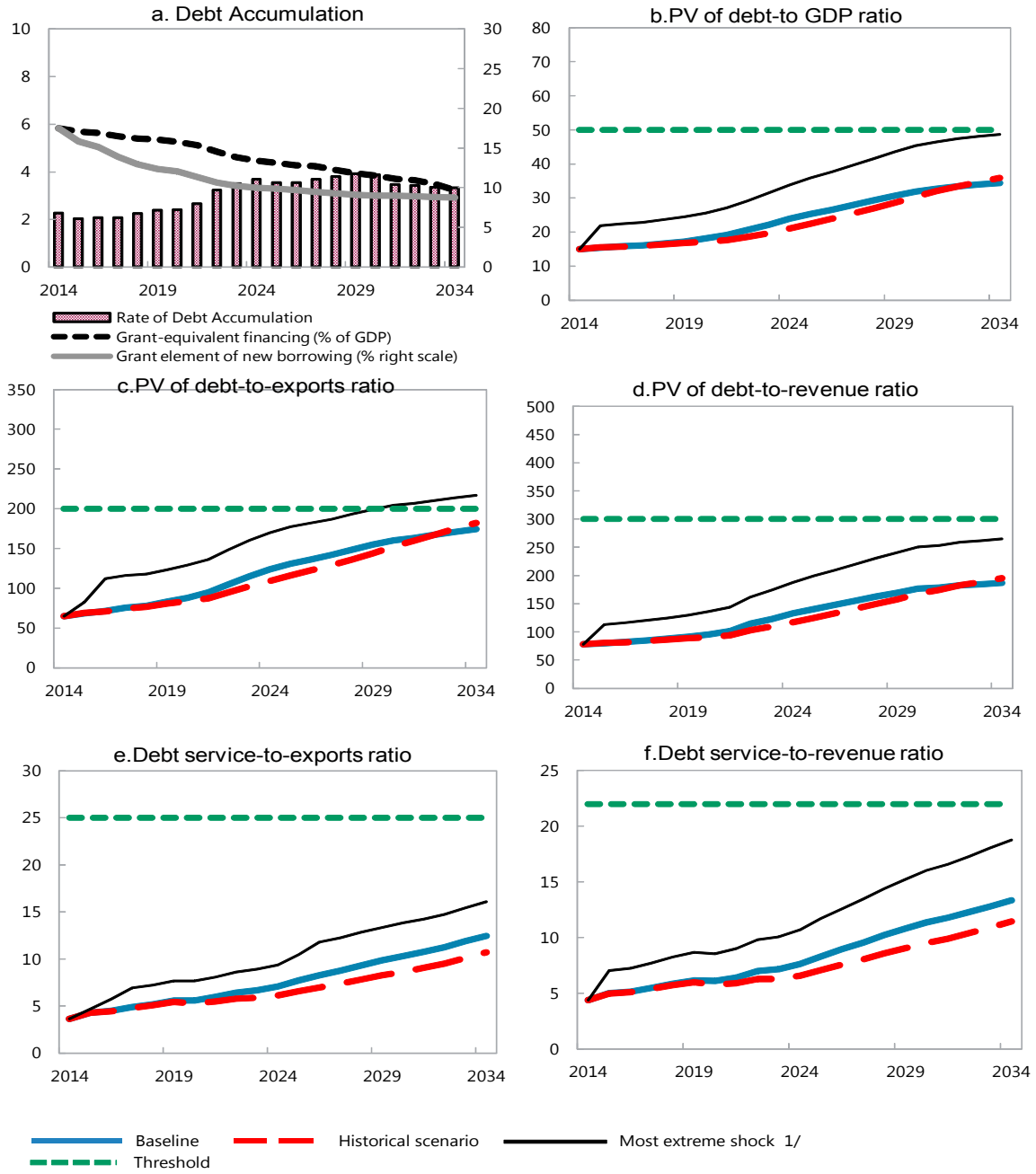
CONCLUSION

11. The DSA results indicate that Burkina Faso's risk of debt distress remains "moderate." The baseline scenario shows no breach of debt distress thresholds for any of the indicators. However, the debt-distress threshold for the PV of debt-to-exports is breached under the standardized stress test of a shock to exports. An alternative low growth scenario combined with the standardized stress tests, shows an earlier breach of this indicator and a breach of the PV of debt-to-GDP.

Authorities' Views

12. The assumptions and conclusions of the DSA were discussed with the authorities who broadly concurred with the assessment and with maintaining a "moderate" debt risk rating. They thought that the alternative scenario of an impact on growth due to a weather shock was useful for considering realistic risks. On domestic debt issuance, they indicated that the new regional debt issuance entity, "Agence UMOA Titres" has been established, and that domestic debt will be issued on variable-bid schemes going forward. The authorities reaffirmed their commitment to seek concessional financing to the greatest extent possible, but they continue to be concerned about the limited availability of such financing for large investment projects.

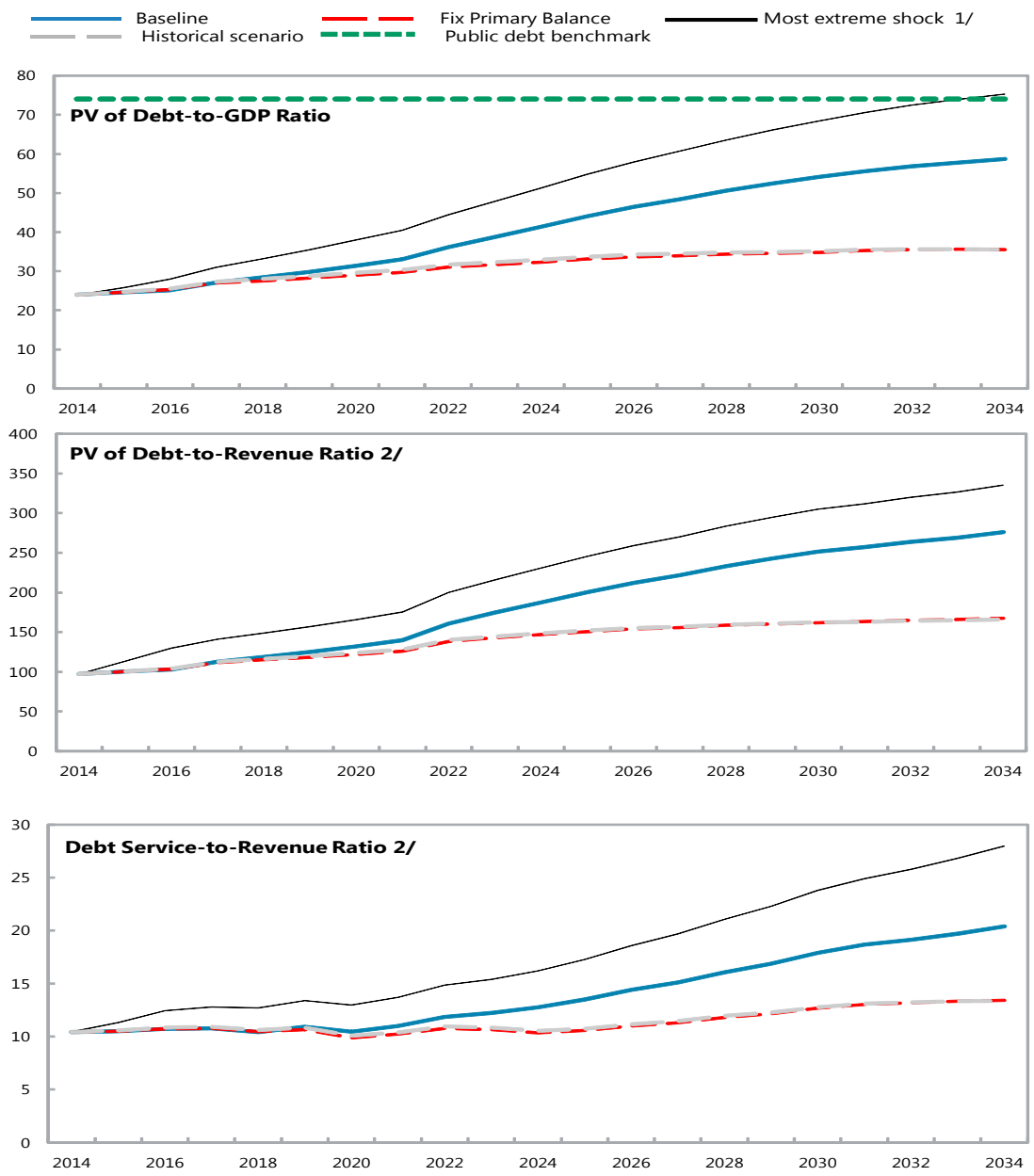
Figure 2. Burkina Faso: Indicators of Public and Publicly Guaranteed External Debt Under Alternatives Scenarios, 2014–34[1]



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a One-time depreciation shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Figure 3. Burkina Faso: Indicators of Public Debt Under Alternative Scenarios, 2014–34[1]



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2024.
2/ Revenues are defined inclusive of grants.

Table 3. Burkian Faso: External Debt Sustainability Framework, Baseline Scenario, 2011–34[1]

(Percent of GDP, unless otherwise indicated

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2011	2012	2013			2014	2015	2016	2017	2018	2019	2014-2019 Average	2024	2034	2020-2034 Average
External debt (nominal) 1/	**24.6**	**22.9**	**21.8**			**22.2**	**21.9**	**21.8**	**21.7**	**21.8**	**22.1**		**28.2**	**38.9**	
of which: public and publicly guaranteed (PPG)	24.6	22.9	21.8			22.2	21.9	21.8	21.7	21.8	22.1		28.2	38.9	
Change in external debt	-1.5	-1.7	-1.1			0.4	-0.2	-0.2	-0.1	0.1	0.3		1.7	0.4	
Identified net debt-creating flows	-2.2	1.0	3.8			4.6	4.5	4.7	4.7	4.8	4.9		4.9	3.2	
Non-interest current account deficit	**1.3**	**4.3**	**6.8**	**7.0**	**3.8**	**6.9**	**6.6**	**6.6**	**6.6**	**6.6**	**6.6**		**6.8**	**5.8**	6.4
Deficit in balance of goods and services	6.9	8.7	10.4			10.6	10.1	10.1	10.0	10.0	10.0		10.1	8.5	
Exports	26.9	26.5	24.6			23.3	22.5	22.1	21.4	21.4	20.8		19.3	19.7	
Imports	33.9	35.1	35.0			33.9	32.6	32.2	31.4	31.3	30.8		29.4	28.2	
Net current transfers (negative = inflow)	-5.5	-4.8	-4.0	-4.9	0.8	-3.8	-3.6	-3.5	-3.4	-3.3	-3.2		-2.8	-1.8	-2.6
of which: official	-4.2	-3.5	-2.7			-2.6	-2.4	-2.4	-2.3	-2.3	-2.2		-1.8	-1.0	
Other current account flows (negative = net inflow)	-0.2	0.4	0.4			0.2	0.1	0.1	0.0	-0.1	-0.1		-0.4	-0.9	
Net FDI (negative = inflow)	**-0.4**	**-2.3**	**-1.3**	**-1.4**	**1.4**	**-1.3**	**-1.2**	**-0.9**	**-1.0**	**-1.0**	**-1.1**		**-1.3**	**-2.0**	-1.6
Endogenous debt dynamics 2/	**-3.1**	**-1.0**	**-1.7**			**-1.0**	**-1.0**	**-1.0**	**-0.9**	**-0.8**	**-0.7**		**-0.6**	**-0.6**	
Contribution from nominal interest rate	0.2	0.3	0.2			0.3	0.4	0.4	0.5	0.5	0.6		0.9	1.5	
Contribution from real GDP growth	-1.1	-2.1	-1.4			-1.3	-1.4	-1.4	-1.3	-1.3	-1.3		-1.5	-2.1	
Contribution from price and exchange rate changes	-2.2	0.9	-0.5			...	...	...	...	...	...		...	...	
Residual (3-4) 3/ with changes in project grants	**2.2**	**-0.7**	**-1.5**			**-0.8**	**-1.2**	**-1.4**	**-1.4**	**-1.2**	**-1.2**		**-0.5**	**-0.7**	
of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	14.1			15.0	15.4	15.8	16.1	16.6	17.2		23.8	34.3	
In percent of exports	...	...	57.4			64.6	68.6	71.6	75.5	77.9	82.8		123.5	174.4	
PV of PPG external debt	**...**	**...**	**14.1**			**15.0**	**15.4**	**15.8**	**16.1**	**16.6**	**17.2**		**23.8**	**34.3**	
In percent of exports	**...**	**...**	**57.4**			**64.6**	**68.6**	**71.6**	**75.5**	**77.9**	**82.8**		**123.5**	**174.4**	
In percent of government revenues	**...**	**...**	**75.1**			**77.9**	**79.9**	**82.1**	**84.9**	**88.0**	**91.5**		**132.6**	**186.7**	
Debt service-to-exports ratio (in percent)	**2.2**	**2.7**	**3.0**			**3.6**	**4.3**	**4.5**	**4.9**	**5.2**	**5.6**		**7.1**	**12.5**	
PPG debt service-to-exports ratio (in percent)	**2.2**	**2.7**	**3.0**			**3.6**	**4.3**	**4.5**	**4.9**	**5.2**	**5.6**		**7.1**	**12.5**	
PPG debt service-to-revenue ratio (in percent)	**3.7**	**4.0**	**4.0**			**4.4**	**5.0**	**5.1**	**5.5**	**5.8**	**6.2**		**7.6**	**13.3**	
Total gross financing need (Billions of U.S. dollars)	0.2	0.3	0.8			0.9	1.0	1.1	1.2	1.3	1.5		2.3	4.6	
Non-interest current account deficit that stabilizes debt ratio	2.8	6.0	8.0			6.6	6.9	6.8	6.7	6.5	6.4		5.2	5.3	
Key macroeconomic assumptions															
Real GDP growth (in percent)	5.0	9.0	6.6	6.1	2.1	6.7	6.8	7.0	6.8	6.7	6.6	6.8	6.3	6.0	6.2
GDP deflator in US dollar terms (change in percent)	9.2	-3.4	2.4	4.9	6.9	4.2	3.8	3.2	3.6	3.4	3.4	3.6	2.0	2.2	2.0
Effective interest rate (percent) 5/	1.0	1.2	1.0	1.0	0.2	1.4	1.8	2.1	2.3	2.7	2.9	2.2	3.8	4.3	4.0
Growth of exports of G&S (US dollar terms, in percent)	46.8	3.7	1.4	26.8	28.2	5.1	7.1	8.4	7.2	10.3	7.4	7.6	8.5	7.7	7.9
Growth of imports of G&S (US dollar terms, in percent)	36.5	9.4	8.8	17.3	15.7	7.4	6.7	9.0	7.9	10.3	8.6	8.3	7.8	6.9	7.7
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	17.5	15.9	15.1	13.9	12.9	12.3	14.6	10.0	8.7	9.8
Government revenues (excluding grants, in percent of GDP)	16.1	17.7	18.8			19.3	19.3	19.3	19.0	18.9	18.8		18.0	18.4	18.2
Aid flows (in Billions of US dollars) 7/	0.8	0.8	0.8			0.8	0.9	0.9	1.0	1.1	1.2		1.4	2.2	
of which: Grants	0.5	0.5	0.7			0.7	0.8	0.9	0.9	1.0	1.1		1.3	2.1	
of which: Concessional loans	0.3	0.2	0.1			0.1	0.1	0.1	0.1	0.1	0.1		0.1	0.1	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			5.8	5.7	5.6	5.5	5.4	5.4		4.5	3.2	4.2
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			72.7	72.9	72.3	71.5	69.2	67.7		54.4	45.0	53.1
Memorandum items:															
Nominal GDP (Billions of US dollars)	10.5	11.0	12.0			13.4	14.8	16.4	18.1	20.0	22.1		33.2	73.4	
Nominal dollar GDP growth	14.6	5.3	9.2			11.1	10.8	10.5	10.7	10.4	10.3	10.6	8.4	8.4	8.3
PV of PPG external debt (in Billions of US dollars)			1.8			2.0	2.3	2.6	2.9	3.4	3.8		8.0	25.4	
(PVt-PVt-1)/GDPt-1 (in percent)						2.3	2.0	2.1	2.1	2.2	2.4	2.2	3.7	3.3	3.4
Gross workers' remittances (Billions of US dollars)	0.2	0.2	0.2			0.2	0.2	0.2	0.2	0.2	0.3		0.4	0.6	
PV of PPG external debt (in percent of GDP + remittances)	...	...	13.9			14.8	15.2	15.6	16.0	16.4	17.0		23.6	34.0	
PV of PPG external debt (in percent of exports + remittances)	...	...	54.1			60.9	64.7	67.7	71.4	73.7	78.3		117.0	166.9	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	2.9			3.4	4.1	4.2	4.6	4.9	5.3		6.7	11.9	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that PV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 4. Burkina Faso: Public Sector Debt Sustainability Framework, Baseline Scenario, 2011–34

(Percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2011	2012	2013	Average 5/	Standard Deviation 5/	2014	2015	2016	2017	2018	2019	2014-19 Average	2024	2034	2020-34 Average
Public sector debt 1/	30.5	28.7	29.1			31.1	31.2	31.2	32.7	33.6	34.7		45.7	63.2	
of which: foreign-currency denominated	24.6	22.9	21.8			22.2	21.9	21.8	21.7	21.8	22.1		28.2	38.9	
Change in public sector debt	1.3	-1.8	0.4			1.9	0.1	0.0	1.5	0.9	1.1		2.6	0.9	
Identified debt-creating flows	0.1	-0.5	1.1			0.8	0.3	0.1	0.7	0.9	1.0		2.4	1.1	
Primary deficit	2.0	2.6	2.5	1.6	6.6	1.7	1.7	1.5	2.0	2.4	2.5	2.0	4.1	3.5	3.7
Revenue and grants	21.2	22.7	24.2			24.7	24.6	24.5	24.2	24.0	23.8		22.0	21.2	
of which: grants	5.1	4.9	5.5			5.4	5.3	5.3	5.2	5.1	5.0		4.0	2.9	
Primary (noninterest) expenditure	23.2	25.3	26.8			26.3	26.3	26.0	26.2	26.4	26.3		26.1	24.8	
Automatic debt dynamics	-1.9	-3.1	-1.5			-0.8	-1.5	-1.4	-1.3	-1.5	-1.5		-1.7	-2.5	
Contribution from interest rate/growth differential	-2.4	-3.2	-0.5			-0.7	-1.1	-1.1	-1.0	-1.3	-1.2		-1.7	-2.5	
of which: contribution from average real interest rate	-1.0	-0.7	1.2			1.1	0.8	1.0	1.0	0.8	0.9		0.9	1.1	
of which: contribution from real GDP growth	-1.4	-2.5	-1.8			-1.8	-2.0	-2.0	-2.0	-2.1	-2.1		-2.5	-3.5	
Contribution from real exchange rate depreciation	0.5	0.1	-0.9			-0.1	-0.3	-0.3	-0.3	-0.3	-0.3		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	1.2	-1.3	-0.6			1.1	-0.2	-0.1	0.8	0.0	0.1		0.2	-0.2	
Other Sustainability Indicators															
PV of public sector debt	...	...	21.5			24.0	24.7	25.2	27.2	28.4	29.8		41.3	58.7	
of which: foreign-currency denominated	...	...	14.1			15.0	15.4	15.8	16.1	16.6	17.2		23.8	34.3	
of which: external	...	...	14.1			15.0	15.4	15.8	16.1	16.6	17.2		23.8	34.3	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	3.3	4.1	4.6			4.3	4.3	4.1	4.6	4.9	5.1		6.9	7.9	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	88.6			97.1	100.3	102.8	112.4	118.6	125.1		187.5	276.3	
PV of public sector debt-to-revenue ratio (in percent)	...	...	114.3			124.1	127.8	130.8	142.8	150.3	158.3		229.7	319.3	
of which: external 3/	...	...	75.1			77.9	79.9	82.1	84.9	88.0	91.5		132.6	186.7	
Debt service-to-revenue and grants ratio (in percent) 4/	6.1	6.6	8.4			10.4	10.5	10.7	10.8	10.4	10.9		12.8	20.4	
Debt service-to-revenue ratio (in percent) 4/	8.0	8.5	10.8			13.3	13.4	13.6	13.7	13.2	13.8		15.6	23.6	
Primary deficit that stabilizes the debt-to-GDP ratio	0.7	4.4	2.1			-0.3	1.6	1.5	0.5	1.5	1.4		1.5	2.6	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	5.0	9.0	6.6	6.1	2.1	6.7	6.8	7.0	6.8	6.7	6.6	6.8	6.3	6.0	6.2
Average nominal interest rate on forex debt (in percent)	1.0	1.2	1.0	1.0	0.2	1.4	1.8	2.1	2.3	2.7	2.9	2.2	3.8	4.3	4.0
Average real interest rate on domestic debt (in percent)	1.7	0.4	15.4	3.6	5.3	15.1	10.5	10.8	10.3	6.6	6.3	10.0	2.8	1.5	2.7
Real exchange rate depreciation (in percent, + indicates depreciation)	2.3	0.5	-4.3	-0.8	8.3	-0.7	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	5.6	4.6	-0.9	3.1	3.1	1.1	1.9	2.0	2.0	2.0	2.0	1.8	2.0	2.2	2.0
Growth of real primary spending (deflated by GDP deflator, in percent)	2.5	18.9	12.9	3.5	6.7	4.9	6.7	5.8	7.6	7.5	6.5	6.5	6.2	5.3	5.7
Grant element of new external borrowing (in percent)	...	...	...	...	...	17.5	15.9	15.1	13.9	12.9	12.3	14.6	10.0	8.7	...

Sources: Country authorities; and staff estimates and projections.

1/ Medium term and long term general government gross debt.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 5. Burkina Faso: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014–34

(Percent)

	Projections							
	2014	2015	2016	2017	2018	2019	**2024**	2034
PV of debt-to GDP ratio								
Baseline	15	15	16	16	17	17	**24**	34
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	15	15	16	16	16	17	**21**	36
A2. New public sector loans on less favorable terms in 2014-2034 2	15	16	17	18	19	20	**30**	48
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	15	16	17	17	18	18	**25**	37
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	15	17	21	21	21	21	**27**	36
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	15	16	18	18	19	19	**27**	39
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	15	16	17	17	18	18	**25**	35
B5. Combination of B1-B4 using one-half standard deviation shocks	15	15	14	14	15	16	**23**	36
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	15	22	22	23	24	24	**34**	49
PV of debt-to-exports ratio								
Baseline	65	69	72	76	78	83	**124**	174
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	65	69	71	74	76	81	**109**	182
A2. New public sector loans on less favorable terms in 2014-2034 2	65	71	77	83	88	96	**156**	243
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	65	69	72	76	78	83	**124**	176
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	65	83	112	116	118	123	**170**	217
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	65	69	72	76	78	83	**124**	176
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	65	72	76	80	82	87	**128**	177
B5. Combination of B1-B4 using one-half standard deviation shocks	65	60	52	56	59	64	**103**	157
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	65	69	72	76	78	83	**124**	176
PV of debt-to-revenue ratio								
Baseline	78	80	82	85	88	92	**133**	187
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	78	80	81	84	86	89	**117**	195
A2. New public sector loans on less favorable terms in 2014-2034 2	78	83	88	94	100	106	**167**	260
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	78	83	87	90	94	97	**141**	199
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	78	89	108	109	111	114	**152**	194
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	78	85	92	95	99	103	**149**	210
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	78	84	88	90	93	96	**137**	189
B5. Combination of B1-B4 using one-half standard deviation shocks	78	77	70	74	78	83	**130**	198
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	78	113	116	120	125	130	**188**	265

Table 5. Burkina Faso: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014–34 (concluded)

(Percent)

Debt service-to-exports ratio								
Baseline	4	4	4	5	5	6	**7**	12
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	4	4	4	5	5	5	**6**	11
A2. New public sector loans on less favorable terms in 2014-2034 2	4	4	4	5	5	6	**8**	18
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	4	4	4	5	5	6	**7**	12
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	4	5	6	7	7	8	**9**	16
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	4	4	4	5	5	6	**7**	12
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	4	4	5	5	5	6	**7**	13
B5. Combination of B1-B4 using one-half standard deviation shocks	4	4	4	4	4	5	**6**	11
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	4	4	4	5	5	6	**7**	12
Debt service-to-revenue ratio								
Baseline	4	5	5	5	6	6	**8**	13
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	4	5	5	5	6	6	**7**	11
A2. New public sector loans on less favorable terms in 2014-2034 2	4	5	5	5	6	6	**9**	19
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	4	5	5	6	6	7	**8**	14
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	4	5	5	6	7	7	**8**	14
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	4	5	6	6	7	7	**8**	15
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	4	5	5	6	6	6	**8**	14
B5. Combination of B1-B4 using one-half standard deviation shocks	4	5	5	5	6	6	**8**	14
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	4	7	7	8	8	9	**11**	19
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	9	9	9	9	9	9	**9**	9

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 6. Burkina Faso: Sensitivity Analysis for Key Indicators of Public Debt, 2014–34

	Projections							
	2014	2015	2016	2017	2018	2019	2024	2034
PV of Debt-to-GDP Ratio								
Baseline	24	25	25	27	28	30	41	59
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	24	25	26	27	28	29	33	35
A2. Primary balance is unchanged from 2014	24	25	25	27	28	28	32	36
A3. Permanently lower GDP growth 1/	24	25	26	28	30	32	47	77
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	24	26	28	31	33	35	51	75
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	24	31	37	38	39	40	50	64
B3. Combination of B1-B2 using one half standard deviation shocks	24	28	32	34	36	38	51	72
B4. One-time 30 percent real depreciation in 2015	24	31	31	32	33	34	45	65
B5. 10 percent of GDP increase in other debt-creating flows in 2015	24	34	34	35	36	37	47	63
PV of Debt-to-Revenue Ratio 2/								
Baseline	97	100	103	112	119	125	188	276
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	97	100	104	113	116	120	148	166
A2. Primary balance is unchanged from 2014	97	100	103	112	115	118	147	167
A3. Permanently lower GDP growth 1/	97	101	104	115	123	132	210	357
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	97	104	113	127	137	147	230	352
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	97	125	151	159	163	168	226	303
B3. Combination of B1-B2 using one half standard deviation shocks	97	113	130	141	149	157	231	335
B4. One-time 30 percent real depreciation in 2015	97	124	125	133	138	143	206	305
B5. 10 percent of GDP increase in other debt-creating flows in 2015	97	137	138	146	151	156	216	296
Debt Service-to-Revenue Ratio 2/								
Baseline	10	11	11	11	10	11	13	20
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	10	11	11	11	11	11	11	13
A2. Primary balance is unchanged from 2014	10	11	11	11	10	11	10	13
A3. Permanently lower GDP growth 1/	10	11	11	11	11	11	14	26
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	10	11	11	12	12	12	15	26
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	10	11	12	15	15	13	15	23
B3. Combination of B1-B2 using one half standard deviation shocks	10	11	12	13	13	13	15	25
B4. One-time 30 percent real depreciation in 2015	10	11	12	13	13	13	16	28
B5. 10 percent of GDP increase in other debt-creating flows in 2015	10	11	13	16	12	13	14	23

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.